

Weekly Internship Report (30/5/26 - 5/6/26)

WEEK 3

Intern's Name: Atharv Gupta

Internship Week: Week 3 (30/5/26 - 5/6/26)

Company/Organization Name: Red Cross Research Foundation

Mentor's Name: Shivam Mittal

Week Overview:

- Learned concepts related to Social Engineering and Dark Psychology.
- Studied the "Problem → Reaction → Solution" manipulation cycle and its role in influence techniques.
- Completed a Network Security Assessment focused on risks in active ports and process communications.
- Performed an Enterprise-Grade Web Application Security Assessment using OWASP ZAP.
- Solved multiple Prototype Pollution labs and Web LLM Attack scenarios.
- Conducted web application penetration testing activities on multiple organizations under supervision and within the internship learning environment.

Learning Outcomes:

- Understood psychological manipulation techniques used in social engineering attacks.
- Learned how prototype pollution vulnerabilities can lead to DOM XSS and client-side attacks.
- Gained practical exposure to OWASP ZAP and web security assessment methodologies.
- Improved understanding of Web LLM attack surfaces and prompt-based exploitation techniques.
- Enhanced skills in vulnerability identification, attack surface analysis, and web application penetration testing.

Challenges Faced:

- Understanding prototype pollution attack chains and exploitation paths.
- Analyzing complex client-side vulnerabilities and browser-based attack vectors.
- Learning advanced web application testing methodologies and security assessment processes.

Goals for Next Week:

- Learn Claude Code and AI Security Automation concepts.
- Explore AI-assisted security workflows and automation techniques.
- Study Identity-Based Attacks and common authentication threats.
- Understand Passwordless Authentication mechanisms and security benefits.

- Improve practical cybersecurity research and documentation skills.

Mentor Feedback and Suggestions:

- Demonstrated good progress in web security testing and vulnerability analysis.
- Continue strengthening practical penetration testing and reporting skills.
- Maintain proper documentation of findings and learning activities.

Additional Notes:

- This week provided significant hands-on exposure to web application security, prototype pollution vulnerabilities, OWASP testing methodologies, and penetration testing practices.

Student Diary (Log) Entry

Week	Task Assigned	Activities Performed	Key Learnings	Additional Remarks
1	Cybersecurity Fundamentals and Research Tasks	Malware analysis, GitHub repository setup, Dridex research, Steganography challenge, Cryptography exercises, AI SOC and ZTNA discussions	Learned malware analysis, hashing vs encryption, cryptographic workflows, GitHub documentation, and cybersecurity research concepts	Improved practical understanding of cybersecurity operations and research methodologies
2	Threat Intelligence, Container Security and Web Security Research	Studied MD5 hash extraction, Threat Intelligence concepts, Docker and Docker Networking, Kubernetes basics, Ransomware workflow, OSINT and CSINT, Threat Intelligence Framework, GHDB research, Burp Suite and FoxyProxy, and performed XXE Injection vulnerability analysis using Burp Suite	Learned threat intelligence concepts including CVE, TTPs, Pyramid of Pain, threat actors, OSINT/CSINT methodologies, container technologies, web vulnerability assessment, and XXE exploitation techniques	Enhanced practical cybersecurity research, threat analysis, reconnaissance, and web application security skills
3	Web Application Security Assessment and Penetration Testing	Network Security Assessment, OWASP ZAP security testing, Prototype Pollution labs, Web LLM Attack labs, and web application penetration testing exercises	Learned prototype pollution, DOM XSS, OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies	OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies Strengthened practical web application security and vulnerability assessment skills